AssistVision: Clip-On Camera Glasses for Before-and-After Meal-Intake Monitoring in Older Adults

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***Abstract*—Many older adults eat less than they are served. In hospital, eating little predicts death even after adjusting for illness, yet intake is usually judged by eye. AssistVision is a clip-on camera for ordinary spectacles (Seeed XIAO ESP32S3 Sense, a small LiPo and two 3D-printed pods, about INR 2,000–2,900 in parts) worn by a carer. One button press records a 13-frame burst while the carer looks across the plate, once when the meal is served and once when it is cleared; the host reconstructs both scenes and reports the share of the meal eaten. A companion app keeps the meal log, alerts family after a run of low-intake meals and prints a weekly report for the doctor. To choose the vision pipeline we test eleven monocular depth models on 507 Nutrition5k dishes and dense multi-view stereo on 22 synthetic plates with exact volume, plus 12 before/after meal pairs seen from a turntable path and from a simulated head sweep. From an overhead view, deployable table-only anchoring of relative models fails (median volume error $\geq$88.4 %), because a fronto-parallel plane cannot fix both scale and shift. Metric models used as-is are off by 8–54$\times$ in volume; fitting their single scale to the table makes ZoeDepth-NYU the best deployable monocular option, at 44.1 % median error. On the 16 textured-food scenes, dense 24-view MVS with marker scale reaches 4.6 % [2.0, 6.1] median absolute error against exact mesh volume, while sparse SfM alone misses 91.3 % and textureless food on a white plate fails (45.1 %). Accuracy collapses below 12 views and under heavy JPEG compression, but not at half resolution. Even sensor-depth volume predicts weighed mass worse than the dataset mean (68.2 vs 52.0 % median error), which is why the rig weighs the plate. No result in this paper comes from the physical rig. On partly eaten plates, the before/after volume difference matches the exact eaten volume to a median 2.8 % [2.0, 7.4] (11/12 within 10 %). On 12 simulated served/cleared pairs seen from the glasses, the eaten share is off by a median of 10.7 percentage points (5/12 within 10 points). A share of the whole plate cannot say that the rice went and the dal stayed, so the app also splits each plate into its dishes from the two photos and turns what was left into protein eaten per day; after three days under 1.0 g per kg body weight, family and doctor are told which dish was left, or that the menu itself is short. On the 6 simulated plates with more than one dish, the dish left most was named correctly each time, and per-dish shares were off by a median of 10.8 points. The glasses have not yet been built or worn, and no clinical or user study has been run.**

I. Introduction

Older adults who eat too little lose weight and muscle and recover more slowly from illness. In the NutritionDay survey of 16,290 hospitalised adults, more than half did not finish the meal they were given, and eating only about a quarter of it went with an adjusted hazard ratio of 2.10 for dying in hospital over the next 30 days (3.02 for eating nothing) [1]. The ESPEN guideline for geriatrics recommends regular malnutrition screening for every older person [2]. Screening and follow-up need a number for how much was eaten. On wards that number usually comes from a nurse's visual estimate of the leftovers; checked against weighing, such estimates hold up on average across a group but are much less reliable for a single patient [3], [4]. At home there is often no record at all, and children living in another city learn of a decline only once it shows as weight loss.

Weighing the plate before and after the meal is the reference method in those validation studies [4]. It is accurate but tedious, so it is rarely done as routine, and a scale at every seat costs money and space. We take the measurement to the person who is already at the table: a carer wearing AssistVision's clip-on camera glasses presses a button at serving and at clearing, the host compares the two scenes, and the companion app shows the share eaten to family members and the doctor.

The camera raises a measurement problem. Perspective projection discards absolute scale: a small plate near the camera and a large plate far away give the same pixels. Metric scale has to come from outside the image: from a reference object, a depth sensor, camera motion with known geometry, or a learned prior. We measure what each choice costs, and whether a short head sweep gives volumes consistent enough for a served/cleared ratio.

Our contributions are:

1) A meal-intake monitoring system for older adults (Sec. IV): served and cleared photo bursts from one wearable camera, a share-eaten rule that flags impossible

pairs, and a companion app with family and clinician views (meal log, two-week trend, low-intake alert, weekly report).
2) A per-dish view of the same two photos (E7): which dish was left, the protein that went with it, and a protein-gap alert that tells the doctor whether the person is leaving the protein dish or the menu has too little.
3) Open, low-cost camera glasses (Sec. V): parametric printable pods for any spectacle frame, burst-capture firmware, a host client and an illustrated build guide.
4) A benchmark of monocular depth models for food on real RGB-D data (Nutrition5k) that keeps deployable and oracle calibration apart.
5) A controlled synthetic study with exact mesh volumes: dense MVS against ground truth, ablations of the factors a device designer controls (views, resolution, compression, marker accuracy, serveware), and partly eaten plates seen from a turntable path (E5) and from a simulated head sweep (E6).
6) A record of design decisions and rejected alternatives (Sec. III), each tied to a measurement or a stated assumption.

II. Related Work

Meal intake in care settings. Hospitals measure intake by weighing plate waste (the reference) or, in routine practice, by visual estimation on quarter or percentage scales. A systematic review of 14 validation studies found visual estimation valid at group level with limited support for individual patients; only two of the studies used modern technology [3]. In a community hospital, visual estimates by nursing assistants differed significantly from weighed intake, with Bland–Altman limits of -121 to $+147$ kcal per tray [4]. Our glasses replace the visual estimate with a measured share; weighing stays the reference. **Food volume.** Nutrition5k [5] provides overhead RGB-D video and weighed dishes; its depth-based volume baseline uses a constant per-pixel footprint at the capture-plane distance. Multi-view food reconstruction has used structure-from-motion with reference cards and dense stereo [6], [7]. **Monocular depth.** Relative depth models (MiDaS [8], Depth Anything [9], [10]) predict depth up to an unknown affine transform of inverse depth; metric models (ZoeDepth [11], Depth Pro [12], metric-finetuned Depth Anything V2) predict absolute depth from priors learned on room- and street-scale scenes. **Reconstruction.** We use COLMAP [13] for SfM and OpenMVS for dense MVS.

III. Design Decisions and Rejected Alternatives

Table I summarises each decision; the evidence column points to the experiment that supports it. Decisions marked *assumption* rest on datasheets or cost quotes, not on our measurements.

IV. Meal-Intake Monitoring for Older Adults

Protocol. When the meal is served, the carer wearing the glasses presses the button and turns their head slowly across the plate; when the plate is cleared they do the same again (Fig. 1; `glasses_client.py meal --phase served|cleared`). Each 13-frame burst is reconstructed on the host into a food volume, V_s at serving and V_l at clearing, and the share eaten is

$$f = 1 - \frac{\min\big(\max(V_l, 0),\, V_s\big)}{V_s}. \tag{1}$$

A served volume under 5 ml is treated as an empty plate. A leftover more than 10 % larger than the serving (food added, a different plate, a failed scan) is stored with a flag and kept out of the alert rule instead of being clipped silently (`intake.photo_intake`).

Why a share, not grams. The glasses carry no scale. Grams from images would need a density for every food, and on real mixed dishes even sensor-measured volume tracks weighed mass poorly (E2). A share needs neither: the plate, the food and its density are the same in both scans, so they cancel in the ratio. This is the quantity nurses already record by eye in quarters of a plate [3], measured instead of guessed. Where the kitchen weighs each portion, grams eaten follow as $f\, m_s$, assuming the leftover has the same density as the serving.

Alerts and reports. A meal is *low* if $f < \tau$; an alert is raised when the last k scored meals are all low, and only once per run. The defaults $\tau = 50\,\%$ and $k = 3$ are prototype values, not clinical cut-offs; the clinician sets both per person. The weekly report lists, per day, the number of meals, the eaten share and the count of low meals. It does not report energy or nutrients.

App. The companion app (Flask and SQLite, Fig. 2) has one page per older adult with the alert banner, weekly figures, two-week chart, recent meals and the circle of care (family members and doctor). Meals arrive through `/api/meals` or are entered by hand. The endpoint takes either served and left grams (the earlier weighing rig) or the two reconstructed volumes from the glasses, which it turns into a share with the rule above and tags as a glasses meal. Running the reconstruction and posting the result is still a manual step on the host. Images stay on the host. In this prototype alerts are recorded in the app but not sent by SMS or e-mail. Unit tests cover the intake arithmetic, the alert rule (including firing once per run) and the client-to-app path.

Which dish was left, and its protein. A share of the whole plate hides the case that matters most on an Indian menu: the rice is finished and the dal is left. The rice gave energy, but most of the protein was in the dal. The host therefore also splits the plate into dishes. It takes the centre frame of each burst, warps it to a top view with the two mat markers, and marks as food every pixel that differs from the bare rim in colour or local texture. The served food is clustered into the K dishes on the menu by colour and position, and each cleared food pixel joins the nearest served dish. A dish's eaten share is $1 - (A_{\text{left}}/A_{\text{served}})^{1.5}$, since a mound that shrinks evenly keeps its volume proportional to its area to the power 1.5. The kitchen enters the protein in one portion of each dish once, from its recipe or the Indian food composition tables,

TABLE I
DESIGN DECISIONS, ALTERNATIVES CONSIDERED, AND THE EVIDENCE BEHIND EACH CHOICE.

Decision	Chosen	Rejected alternatives	Why / evidence
Scale source	Printed ArUco mat, multi-view triangulation	Credit-card reference in one image; metric monocular depth; depth sensor	Volume scales with the *cube* of length error (Sec. VI-B); metric networks are off by large factors at plate distance (Table II); a D435-class sensor costs $\sim$10$\times$ the camera (assumption).
Geometry	Dense MVS (COLMAP + OpenMVS)	Sparse SfM points only; single-view depth	Sparse points under-cover low-texture food (Sec. VII-C); single view cannot see the far side of a mound even with perfect depth.
Capture motion	Head sweep: 13-frame burst from clip-on glasses	Fixed camera on a motorised turntable (our first design); handheld phone orbit; commercial smart glasses	The carer is already at the table, so nothing is added to the tray or table; a 60° sweep gives enough baseline for MVS (E6); the turntable needs mains power, space and a plate placed on it; commercial glasses are closed and cost over 10$\times$ more (assumption).
Camera tilt	Pod tilted 20° down; plate seen at about 55° elevation	Straight overhead (Nutrition5k)	Overhead view makes the support plane fronto-parallel: the relative-depth anchor fit becomes rank-deficient (condition number 10^{14}, Sec. VI-C).
Controller	XIAO ESP32S3 Sense (camera, 8 MB PSRAM, LiPo charger on board)	ESP32-S3 dev board + separate charger; Raspberry Pi Zero + camera; smartphone	Thumb-sized and light enough for a temple arm; no extra charger board or OS; processing is host-side, so the MCU only needs camera, Wi-Fi and a button (assumption: cost table in the build guide).
Burst storage	13 JPEGs held in PSRAM, pulled by the host	Stream video; store on microSD	Frames stay sharp at UXGA instead of video bit rates; no card to remove; the host checks every frame before filing it.
Volume integral	Height field over the fitted support plane	Constant per-pixel footprint (Nutrition5k); voxel carving; mesh + watertight closure	Exact on tilted planes in our unit tests (0.09 % error) where the constant-footprint formula is not; shared across all methods so comparisons measure geometry, not bookkeeping.
Intake measure	Share eaten, $1 - V_l/V_s$, from the two reconstructions	Staff visual estimate; grams via volume $\times$ density; plate scale	The plate, food and density cancel in the ratio; visual estimates are weak for individuals [3]; volume tracks mass poorly (E2); weighing is the reference [4] but needs a scale at every seat.
Alert rule	k low meals in a row, threshold set by the clinician	Daily-total threshold; fixed population cut-off	A run of low meals is visible within a day; thresholds are prototype defaults, not validated cut-offs.
Mono calibration	Report raw / anchor / oracle separately	Report only oracle-aligned numbers	Oracle alignment uses the ground truth being evaluated; it is an upper bound, not a deployable result.

and the app multiplies. The daily total is compared with 1.0 g per kg of body weight, the low end of the 1.0–1.2 g/kg that the PROT-AGE group recommends for healthy older adults [14]; the doctor can change it. After three short days in a row the family and the doctor get one message. If a dish was left (under 75 % of it eaten on average) the message names it, e.g. "Most often left: Sambar". If almost everything was eaten and protein was still short, the message says the menu is short instead, which points at the kitchen rather than the appetite. The low-intake alert cannot separate these cases: in the demo data one resident eats most of every meal, so her low-intake alert never fires, yet she stays under her protein target and the protein alert does. Photographs of plate waste are an established way to estimate what was eaten [15]; here the photos come from the carer's glasses and the result goes straight to the people who can act.

V. HARDWARE

The capture device is a clip-on camera for an ordinary spectacle frame (Fig. 4). A Seeed XIAO ESP32S3 Sense (OV2640, 8 MB PSRAM, 21×17.8 mm) sits in a printed pod that hooks over the temple arm; a single-cell 3.7 V LiPo sits in a second pod behind it and is charged through the board's own USB-C port, so there is no separate charger board. The only wiring is the two battery leads and an optional push button on D1; the board's BOOT button works as the trigger without it. The two pods weigh about 4.6 g and 5.5 g of solid PLA by CAD volume and print without supports. The CAD

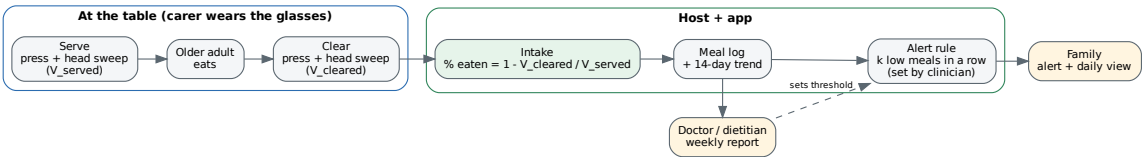


Fig. 1. Meal-intake loop. The glasses record a served and a cleared burst; the host reconstructs both and computes the share eaten; family members get alerts and a daily view; the doctor gets a weekly report and sets the alert threshold.

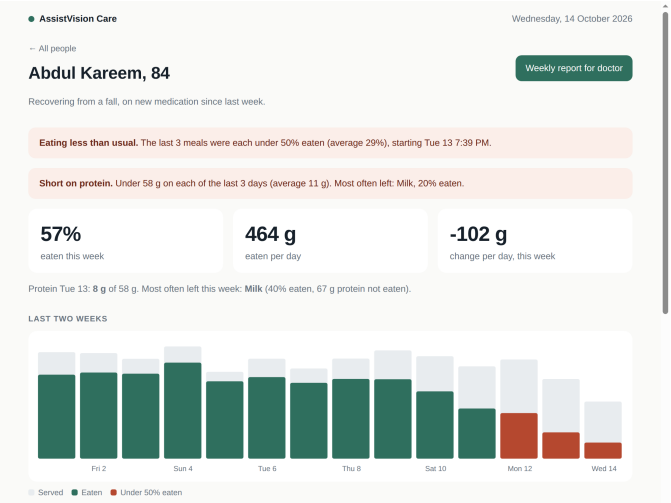


Fig. 2. Companion app, resident page (fictional demo data): low-intake and protein-gap alerts (the second names the dish most often left), weekly share of served food eaten, grams per day, protein on the last full day against the target, and the two-week trend of served (grey) versus eaten (green; red when under the threshold).

is parametric, so a different temple-arm thickness or height means editing two numbers and re-exporting the STL files.

One press records a burst of 13 JPEG frames at 1600×1200, 350 ms apart, into PSRAM while the wearer turns their head slowly across the plate, about 4 s in total. The firmware exposes `/status`, `/trigger`, `/frame?i=`, `/capture` and `/camera`; the host client (`glasses_client.py meal --phase served|cleared`) waits for the burst, checks that every frame is a JPEG, and files it under the meal. It refuses a *cleared* burst for a meal with no *served* burst, and a second burst for the same phase. All reconstruction runs on the host. The firmware compiles for the XIAO ESP32S3 target (Arduino core 3.3.12) and the client passes tests against a simulated device; the glasses have not yet been printed, flashed or worn, so no physical number is reported for them.

Earlier design. We first designed the capture side as a fixed ESP32-S3 camera over a stepper turntable with a load cell under the plate; its firmware and client remain in the repository. E3–E5 use that design's 24-view turntable path; E6 repeats the meal pairs from the glasses' viewpoint. Neither device has been measured physically yet.

VI. Method

A. Shared volume integral

All methods produce a 3-D point set P in metres and a support plane $(\mathbf{n}, d)$ oriented towards the camera. Each point gets a height $h = \mathbf{n}^\top \mathbf{p} + d$ and plane coordinates in an orthonormal basis; points are binned into square cells of side c and each cell keeps the median height. Empty cells inside the footprint are filled by nearest neighbour, negative heights are clipped, and

$$V = c^2 \sum_{\text{cells}} \max(h, 0). \tag{2}$$

Because monocular, multi-view and sensor depth all share this integral, differences in volume reflect differences in geometry, not in post-processing.

B. Metric scale from the marker mat

Marker corners are detected in every view and triangulated with the SfM cameras; the scale s is the ratio of the known side to the median triangulated side. A relative error ϵ in s produces a volume error of $(1+\epsilon)^3 - 1 \approx 3\epsilon$, so a 1 % scale error costs about 3 % in volume; Sec. VII-C measures this directly.

C. Monocular protocols

Raw: the model's metric output is used as-is. **Anchor**: the prediction is calibrated using only the support plane (the table in Nutrition5k, the marker squares beside our plate), never the food; metric outputs get one scale, relative outputs an affine map of inverse depth. **Oracle**: the same maps fitted to ground-truth depth over the whole dish region—an upper bound that is not available at deployment. For relative models we record the condition number of the anchor fit; a fronto-parallel plane has nearly constant inverse depth, so scale and shift are not separately identifiable.

VII. Experiments

A. E1: Monocular depth on Nutrition5k

Relative models: anchoring on the table fails from overhead. The five relative models have a median volume error of 88.4–99.4 % under the deployable anchor protocol (Table II), even though their predicted table is flat to 0.6–0.7 mm RMS: the scale-and-shift fit reproduces the table and collapses the food relief. The fit is badly conditioned on most

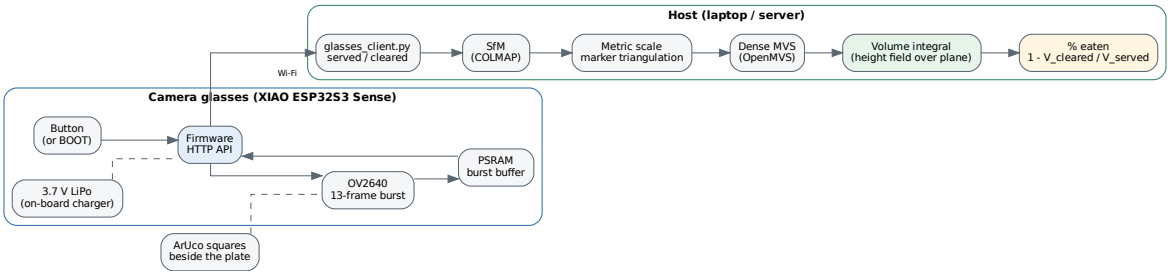


Fig. 3. System architecture. The glasses only capture; the host pulls each burst over Wi-Fi, reconstructs it and computes the share eaten.

TABLE II
E1 (NUTRITION5K, n DISHES PER MODEL AS SHOWN). DEPTH ERROR OVER THE DISH REGION AND VOLUME ERROR RELATIVE TO SENSOR-DEPTH VOLUME, PER MODEL AND CALIBRATION PROTOCOL. BRACKETS: 95% BOOTSTRAP CI (10 000 RESAMPLES). LATENCY IS CPU TIME PER IMAGE ON 8-VCPU CLOUD VMS (VM EVIDENCE; SEE TEXT).

Model	Prot.	AbsRel	h MAE (mm)	Vol. APE med. (%)	$\leq$20% (%)	APE, sens. plane (%)	table RMS (mm)	s/img
DepthPro	raw	0.201	19.0	97.9 [79.4, 100.0]	12	100.0	9.2	40.08
DA-V2-Metric-Indoor-L	raw	1.546	58.3	759.3 [627.7, 860.3]	2	100.0	26.7	3.80
DA-V2-Metric-Indoor-B	raw	2.028	118.2	772.2 [356.3, 1090.1]	2	100.0	51.5	2.18
DA-V2-Metric-Indoor-S	raw	1.334	55.8	801.3 [645.5, 942.5]	3	100.0	25.4	1.70
ZoeDepth-NYU-KITTI	raw	2.759	79.0	3539.3 [3267.7, 3906.8]	0	100.0	34.9	2.76
ZoeDepth-NYU	raw	2.613	86.2	5327.9 [5074.6, 5574.9]	0	100.0	34.1	2.64
ZoeDepth-NYU	anchor	0.049	18.5	44.1 [39.6, 50.5]	24	49.4	9.4	2.64
DepthPro	anchor	0.037	14.7	47.7 [41.7, 54.9]	24	47.4	7.9	40.08
ZoeDepth-NYU-KITTI	anchor	0.050	21.4	58.6 [54.3, 62.8]	18	53.2	9.3	2.76
DA-V2-Metric-Indoor-S	anchor	0.063	24.4	73.5 [67.3, 80.3]	12	70.7	11.1	1.70
DA-V2-Metric-Indoor-L	anchor	0.057	25.8	82.7 [77.7, 86.4]	12	75.6	10.5	3.80
DA-V2-L	anchor	0.035	12.5	88.4 [86.2, 90.1]	6	76.3	0.7	3.94
DA-V2-B	anchor	0.038	13.4	92.7 [90.1, 95.4]	4	84.9	0.6	2.89
DA-V1-S	anchor	0.039	13.6	94.8 [92.9, 96.0]	3	86.1	0.6	1.64
DA-V2-S	anchor	0.038	13.4	96.0 [93.4, 97.8]	3	86.9	0.6	1.85
MiDaS-DPT-Hybrid	anchor	0.045	15.8	99.4 [98.9, 99.8]	1	93.7	0.7	1.39
DA-V2-Metric-Indoor-B	anchor	0.125	45.8	99.7 [98.7, 99.9]	7	99.6	18.3	2.18
DA-V2-L	oracle	0.013	11.2	25.9 [23.7, 30.2]	38	9.6	3.0	3.94
DA-V2-B	oracle	0.015	11.8	32.1 [28.5, 35.7]	29	13.0	2.7	2.89
DA-V2-S	oracle	0.016	12.0	37.4 [33.0, 41.9]	26	14.1	2.6	1.85
ZoeDepth-NYU	oracle	0.036	20.0	45.1 [40.7, 50.8]	23	33.3	9.5	2.64
DA-V1-S	oracle	0.017	10.3	47.6 [42.9, 51.8]	17	17.2	2.4	1.64
DepthPro	oracle	0.028	15.1	49.5 [43.5, 57.7]	23	30.5	7.9	40.08
ZoeDepth-NYU-KITTI	oracle	0.037	22.5	60.9 [55.7, 65.2]	18	31.6	9.3	2.76
DA-V2-Metric-Indoor-S	oracle	0.042	25.3	75.2 [67.1, 80.7]	12	44.3	10.9	1.70
DA-V2-Metric-Indoor-L	oracle	0.037	25.9	85.5 [80.8, 89.2]	11	41.6	10.4	3.80
MiDaS-DPT-Hybrid	oracle	0.022	11.4	86.9 [82.7, 89.0]	3	33.8	1.1	1.39
DA-V2-Metric-Indoor-B	oracle	0.068	42.7	99.8 [99.2, 100.0]	6	67.3	17.5	2.18

dishes (Fig. 5), as expected when the only reference is a plane almost perpendicular to the optical axis. This is the quantitative reason the rig views the plate at 45° and uses the marker mat and multiple views.

Metric models are not metric at plate distance, but one scale helps. Used as-is, all five metric models are far off: AbsRel 1.33–2.76 and median volume errors of 759–801 % (DA-V2 metric indoor) and 3,539–5,328 % (ZoeDepth). ZoeDepth places the table at about 1.2 m instead of 0.35 m, a distance well outside the room-scale scenes it was trained on. Their anchor fits only one scale to the table, which is well conditioned even from overhead. ZoeDepth-NYU is then the best deployable model: 44.1 % [39.6, 50.5] median volume error, 24 % of dishes within 20 %, ahead of the next model (DA-V2-Metric-Indoor-S, 73.5 %; Holm-corrected Wilcoxon $p = 3.6 \times 10^{-9}$). That is still an order of magnitude worse than multi-view reconstruction (Sec. VII-C). Its predicted table is not flat (9.4 mm RMS), and the KITTI-mixed variant is worse (58.6 %). Model size does not order the DA-V2 metric models: B is the worst of the three under every protocol.

Depth Pro is the exception, but no better once anchored. Depth Pro was run on a seeded 200-dish subset because it takes 40 s per image on the VM CPU. Used as-is it is the only metric model with roughly the right scale (AbsRel 0.20), yet it still overestimates volume (median signed error +82 %, absolute 97.9 %). Anchored, it reaches 47.7 % [41.7, 54.9],



Fig. 4. Camera glasses: printed camera pod (XIAO ESP32S3 Sense behind a square window, tilted 20° down) and battery pod clipped to the right temple arm of an ordinary frame. Rendered from the printable CAD files; the frame is a generic stand-in.

indistinguishable from ZoeDepth-NYU on the same dishes (45.3 %; Wilcoxon $p = 0.8$, $n = 198$) at about 15× the latency. Its oracle fit (49.5 %) is no better than its anchor and far behind DA-V2-L ($p = 9.2 \times 10^{-11}$).

Depth metrics do not predict volume. After alignment AbsRel is 0.013–0.125 for every model, yet volume errors span 10–100 %. Food relief (1–3 cm) is a small fraction of the 36 cm camera distance, so a model can score well on depth and still lose most of the volume. We therefore rank models on volume, not on AbsRel.

Oracle upper bound. With an oracle fit DA-V2-L is best of all eleven models: 25.9 % [23.7, 30.2] median volume error with its own support plane and 9.6 % with the sensor's plane. Within Depth Anything V2 (relative) the error falls with encoder size, from 37.4 % (S) to 32.1 % (B) to 25.9 % (L); L is separated from B with Holm-corrected $p = 3.1 \times 10^{-19}$. The gap between the two plane choices shows that fitting the support plane in predicted depth causes more than half of the error; in the rig the marker mat provides this plane independently of the network. The oracle needs food ground truth and is not deployable.

Coverage and failures. 498 of the 503 usable dishes were evaluated for every model except Depth Pro (198 of its 200-dish subset). The remaining five have no visible table (degenerate plane fit) and are counted as failures, not dropped. Latency is 1.4–3.9 s per image on 8-vCPU cloud VMs (VM evidence), and 40 s for Depth Pro: DA-V1-S, DA-V2-S/B and MiDaS ran on a machine shared with other jobs, so their times are upper bounds, and the other models ran alone on identical VMs.

B. E2: Sensor-referenced volume and weighed mass

Nutrition5k has no food-volume label, so the volume target in E1/E2 is the volume integrated from RealSense depth over the dish region with the same height-field integral (Eq. 2) applied to each model's prediction. The error therefore isolates

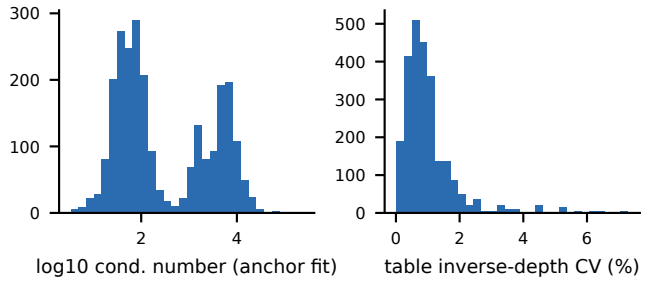


Fig. 5. Anchor-fit conditioning on Nutrition5k (relative models): condition number of the affine fit (left) and coefficient of variation of table inverse depth (right). A low CV means the table alone cannot separate scale from shift.

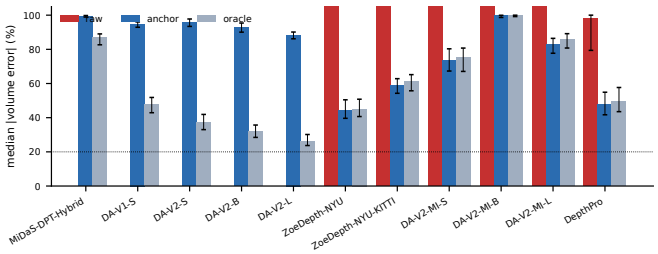


Fig. 6. Per-dish absolute volume error versus sensor depth, by model and protocol (Nutrition5k).

the depth model: a model with zero error here reproduces the sensor, whatever the sensor's own error on food is. Even sensor-depth volume with a cross-validated density predicts mass worse than ignoring the image and predicting the mean (Table III). Sensor volume and weighed mass are only weakly correlated (Pearson $r = 0.19$, Spearman $\rho = 0.17$, $n = 503$). Mixed dishes differ in density, and the dish region includes the plate rim. A single image-derived volume therefore cannot stand in for grams on real dishes, and this is the quantitative case for weighing the plate in the rig. Mass counts n vary because dishes with a non-positive predicted volume are excluded from the density fit.

C. E3: Multi-view versus monocular on exact synthetic volumes

Scenes. We specify twenty-two Blender (Cycles) scenes with watertight food meshes whose volume is computed exactly from the mesh: eight food configurations (single rice mound, rice + dal, two idli, curry heap, lumpy rice, a three-item thali, a 40 mL small portion and a tall mound) on ceramic and polished steel plates (16 scenes), four textureless-food variants of the first four configurations on ceramic, and two configurations on banana leaf. All 22 were rendered and reconstructed. Each scene is rendered at 1600×1200, 60° horizontal FoV, 45° elevation and 0.40 m distance, with the plate rotating on a turntable inside a printed 50 mm ArUco mat, i.e. the rig geometry. View 0 also stores a metric depth map for a "perfect single-view depth" baseline.

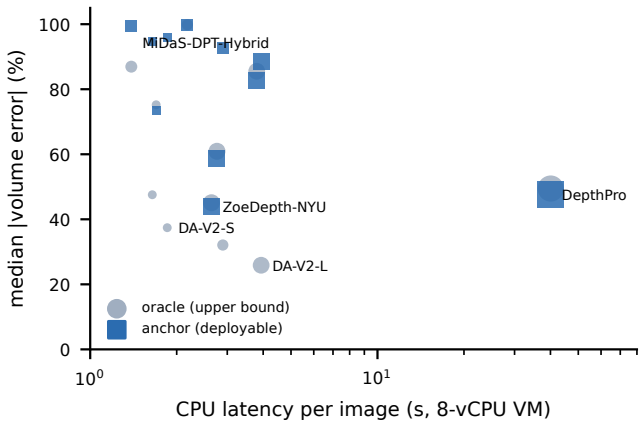


Fig. 7. Accuracy–latency trade-off (anchor and oracle protocols; CPU latency on 8-vCPU cloud VMs).

TABLE III
E2 (NUTRITION5K). MASS PREDICTED AS ρV WITH A SINGLE DENSITY ρ FITTED ON FOUR FOLDS AND SCORED ON THE HELD-OUT FOLD (5-FOLD CV). BRACKETS: 95 % BOOTSTRAP CI. †200-DISH SEEDED SUBSET.

Volume source	n	MAE (g)	median APE (%)
RealSense depth (sensor ceiling)	503	139.3 [129.3, 150.0]	68.2 [63.1, 72.9]
DepthPro†, anchor	191	184.3 [159.0, 212.8]	79.1 [73.3, 83.7]
DA-V2-B, oracle	485	177.2 [164.4, 191.1]	83.8 [82.0, 86.1]
DepthPro†, oracle	191	199.1 [172.5, 228.6]	84.4 [79.0, 87.1]
DA-V2-S, oracle	483	178.1 [161.6, 196.6]	85.7 [84.2, 87.5]
DA-V1-S, oracle	491	180.4 [167.2, 194.1]	88.2 [85.7, 89.6]
DA-V2-L, oracle	486	185.3 [172.1, 199.0]	89.0 [87.5, 90.1]
ZoeDepth-NYU, anchor	487	186.8 [173.3, 201.0]	89.1 [87.1, 91.5]
DA-V2-Met-S, anchor	475	182.3 [168.4, 197.0]	92.2 [89.7, 94.0]
Zoe-NYU-KITTI, anchor	480	195.8 [181.8, 210.0]	95.1 [94.4, 96.2]
ZoeDepth-NYU, oracle	487	197.3 [184.0, 211.2]	95.5 [94.9, 96.3]
DA-V2-L, anchor	491	185.6 [171.0, 200.9]	96.3 [95.4, 97.7]
MiDaS-DPT-Hybrid, oracle	490	193.2 [178.7, 208.5]	96.9 [95.9, 97.6]
DA-V2-Met-S, oracle	475	195.2 [180.5, 211.5]	96.9 [95.8, 97.6]
DA-V2-Met-B, oracle	419	186.2 [171.0, 201.6]	97.0 [95.2, 98.3]
Zoe-NYU-KITTI, oracle	480	200.0 [186.2, 214.1]	97.6 [97.2, 98.0]
DA-V2-Met-L, anchor	473	198.4 [184.7, 212.8]	97.8 [96.9, 98.4]
DA-V2-B, anchor	492	186.9 [172.8, 202.0]	97.9 [96.5, 98.8]
DA-V2-Met-L, oracle	473	198.8 [185.1, 213.2]	98.1 [97.3, 98.7]
DA-V2-S, anchor	490	189.0 [175.8, 202.7]	98.7 [97.9, 99.3]
DA-V2-Met-B, anchor	421	192.8 [177.6, 208.4]	98.8 [97.8, 99.3]
DA-V1-S, anchor	487	201.6 [186.4, 218.5]	99.5 [99.4, 99.7]
MiDaS-DPT-Hybrid, anchor	441	194.0 [180.1, 208.4]	99.9 [99.8, 100.0]
Predict the mean mass (no image)	503	117.9 [109.8, 126.5]	52.0 [46.1, 57.2]

Methods. We compare two pipelines. Dense MVS runs COLMAP SfM, takes scale from the triangulated marker corners, densifies with OpenMVS, cuts the food footprint at a threshold set from plate noise, and integrates. Sparse SfM skips densification and integrates the SfM points with the same scale. Single-view runs on these scenes are in the code but were not run, so the monocular comparison rests on E1.

Footprint rule. Dense MVS leaves millimetre-level noise on the plate surface. A cell counts as food if its height exceeds $\mu + 2\sigma$ of the plate annulus heights. This rule was fixed on `rice_mound__ceramic` before the other scenes were run, and those scenes are held out. A hysteresis variant (seed at 2σ, grow connected cells down to 0.5σ) was added *after* the curry failure below, and is reported as post hoc.

Table IV gives the results. With the pre-registered footprint rule, dense MVS has a median absolute error of 4.6 % [2.0, 6.1] on the 16 textured-food scenes, and 13 of 16 are within 10 %. Steel plates (3.0 %, 8 of 8 within 10 %) do better than ceramic (5.9 %, 5 of 8). Banana leaf gives -1.9 and $+7.8$ %. The textureless white ceramic gives plate-surface noise of 1.9–7.4 mm, against 0.9–1.2 mm on the textured steel. The three textured-food misses are all on ceramic: the curry heap (-26.0 %, below), the 38 mL small portion (-30.1 %; 4.5 mm plate noise is a large share of an 18 mm-high portion) and the tall mound ($+16.8$ %).

Textureless food fails. With the food texture removed, dense MVS gives -1.4 % (curry), -20.8 % (idli), -69.4 % (rice + dal) and -82.9 % (rice mound), a median of 45.1 %. Without texture on the food, the matcher has nothing to triangulate on its surface. Real rice and dal are textured, but smooth sauces and plain white foods are not, so this is the main limit of a passive-camera rig.

Failure case. `curry_heap__ceramic` is -26.0 %: its 6.2 mm plate noise raises the threshold above the low outer skirt of the heap, which is then cut from the footprint. The failure is mostly but not always repeated. Five runs of the identical input on this machine gave -26.0, -26.0, -24.8, -23.1 and $+0.2$ %, and one run on a second VM gave $+0.1$ %. OpenMVS is not deterministic on the textureless plate, and the noise estimate, and with it the threshold, moves between runs. The post-hoc hysteresis rule recovers it (-1.0 %) and lowers the textured-food median error to 2.2 %, but its bias is $+5.5$ % and it overestimates the tall mound on ceramic by 46.1 %, lumpy rice by 24.2 % and the thali by 17.9 %, where plate noise connects to the food. No single height threshold works on every scene. We keep the pre-registered rule as primary and conclude that *separating food from plate*, not depth accuracy, is now the dominant error of the multi-view pipeline. Image-space food segmentation projected onto the height field is the natural next step. **Sparse is not enough.** Sparse SfM points alone have a median error of 91.3 % [79.5, 99.5] over all 22 scenes, nearly always an underestimate: feature points concentrate on the textured mat and food edges, leaving most of the food surface empty. Two steel scenes happen to land within 10 % (thali -0.8 %, tall mound $+3.4$ %), but the steel median is still 78.9 %. Densification is therefore required, and it is the main cost of a scan. Dense reconstruction takes about 119 s per scene, against 18 s for sparse on the VM CPU (VM evidence).

D. E4: Ablations

The ablations change one factor at a time and use the dense MVS pipeline, except the marker-scale ablation, which used sparse SfM: number of views (6, 8, 12, 16, 24, evenly spaced), image resolution (1600×1200, 800×600, 640×480, the OV2640 UXGA, SVGA and VGA modes), JPEG re-encoding with OpenCV at quality 30 and 10 on its 0–100 scale (this is not the OV2640 0–63 quality register, and we did not measure how the two map), a deliberate error in the printed marker size (± 1, ± 2 %), plate material, and three

TABLE IV
E3 ON SYNTHETIC PLATES WITH EXACT MESH VOLUME. ERROR COLUMNS GIVE THE MEDIAN ABSOLUTE ERROR WITH A 95% BOOTSTRAP CI ACROSS SCENES; BIAS IS THE MEAN SIGNED ERROR. TIME: CPU SECONDS PER SCENE ON OUR VM.

Method	Subset	n ok	$\|e\|$ % median [95% CI]	bias %	<10%	s
MVS dense (24 views)	all	22/22	5.0 [2.3, 7.8]	-10.8	16	119
MVS dense (24 views)	textured food	16/16	4.6 [2.0, 6.1]	-4.3	13	120
MVS dense (24 views)	ceramic	8/8	5.9 [2.7, 26.0]	-5.9	5	117
MVS dense (24 views)	steel	8/8	3.0 [0.2, 5.1]	-2.7	8	123
MVS dense (24 views)	banana leaf	2/2	4.8 [1.9, 7.8]	+2.9	2	111
MVS dense (24 views)	ceramic, textureless food	4/4	45.1 [1.4, 82.9]	-43.6	1	113
MVS dense, hysteresis (post-hoc)	all	22/22	2.7 [1.1, 8.8]	-0.7	16	119
MVS dense, hysteresis (post-hoc)	textured food	16/16	2.2 [0.7, 5.0]	+5.5	13	120
MVS dense, hysteresis (post-hoc)	ceramic	8/8	5.5 [2.4, 24.2]	+11.5	5	117
MVS dense, hysteresis (post-hoc)	steel	8/8	0.6 [0.1, 2.0]	-0.4	8	123
MVS dense, hysteresis (post-hoc)	banana leaf	2/2	5.3 [1.9, 8.8]	+3.4	2	111
MVS dense, hysteresis (post-hoc)	ceramic, textureless food	4/4	25.6 [1.0, 58.4]	-27.7	1	113
SfM sparse (24 views)	all	22/22	91.3 [79.5, 99.5]	-81.4	2	18
SfM sparse (24 views)	textured food	16/16	84.5 [75.4, 94.8]	-75.3	2	20
SfM sparse (24 views)	ceramic	8/8	90.0 [70.2, 99.6]	-85.7	0	20
SfM sparse (24 views)	steel	8/8	78.9 [3.4, 99.3]	-65.0	2	21
SfM sparse (24 views)	banana leaf	2/2	93.2 [86.5, 100.0]	-93.2	0	16
SfM sparse (24 views)	ceramic, textureless food	4/4	100.0 [98.4, 100.0]	-99.6	0	15

repeated runs on identical input to measure the run-to-run spread of SfM/MVS, which are not bit-deterministic. A run fails if SfM returns no model, if no marker side is triangulated, or if densification errors; failures are counted, not dropped. The driver (`experiments/synth/run_suite.py`) and the analysis script are in the repository. **Marker-scale error (60 runs, sparse SfM).** We scaled the believed marker side by 0.98–1.02 on 12 scenes. The recovered metric scale tracks the injected factor, but the integrated volume does not change monotonically (mean 71.7, 90.3, 77.9, 66.9 and 47.7 mL for factors 0.98 to 1.02). Two sparse runs of the same input differ by a median of 1.9 and up to 62.9 percentage points of true volume, so run-to-run SfM variation is far larger than the $\pm 6\,\%$ that the cube law predicts for a $\pm 2\,\%$ scale error. The cube law is therefore an analytic bound (Sec. VI-B), and this ablation is uninformative on sparse reconstructions. It will be repeated on dense reconstructions.

View count (60 dense runs, 12 scenes). Accuracy collapses below 12 views. The median absolute volume error is 91.6 % [78.7, 96.3] at 6 views and 71.4 % [22.7, 76.5] at 8 views (0 and 2 of 12 scenes within 10 %). It is 3.3 % [0.4, 19.2] at 12, 1.6 % [0.4, 3.6] at 16 and 4.2 % [0.4, 5.6] at 24 views (9, 11 and 12 of 12 within 10 %). The paired difference between 16 and 24 views is not significant ($p = 0.38$, Wilcoxon, $n = 12$). Every view registered at every setting; what drops is dense coverage (median 0.17 M points at 6 views against 1.03 M at 24). Four idli runs at 6 and 8 views return fewer than 1,000 food points and $-100\,\%$ volume while reporting success, so the pipeline needs a minimum-food-point check before a scan is accepted. We keep 24 views as the default because it is the only setting with all scenes within 10 %; 16 views is the fallback when scan time matters. These runs were split across three identical cloud VMs (8-vCPU Xeon, same package versions). The pipeline is not bit-reproducible: the same 24-view input reconstructed on two machines differed by a median of 0.9 percentage points, 10 of 12 scenes within 1.2 points, but by 4.9 points on `thali3__ceramic` and 26.1 points on `curry_heap__ceramic` ($-26.0\,\%$ vs. $+0.1\,\%$).

Resolution and compression (72 dense runs, 12 scenes). Resolution matters less than compression. Median absolute volume error is 3.9 % [0.8, 5.1] at 1600×1200 lossless and 3.2 % [2.1, 4.0] at 800×600 lossless (12 of 12 within 10 %; paired $p = 0.27$). OpenCV quality 30 at full resolution is 7.4 % [2.4, 8.6] (12 of 12 within 10 %, $p = 0.18$). Quality 10 fails at every resolution: 72.7 % at 1600×1200, 90.4 % at 800×600 and 82.7 % at 640×480 (at most 1 of 12 within 10 %; $p = 4.9 \times 10^{-4}$ for each against the lossless baseline, significant after Holm correction over five comparisons). Heavy compression removes the fine texture that dense matching needs. For the rig this means a low compression setting on the camera matters more than its maximum resolution; we did not measure how the OV2640 quality register maps onto these OpenCV levels, so the rig default has to be checked on hardware.

Repeat runs (18 dense runs, 6 scenes, one machine). Three runs on identical input stay within 0.2–0.9 percentage points of each other on steel plates and within 1.8–2.9 points on ceramic. The spread is again larger on the textureless plate. Two observations are already available from E3. Ceramic and steel differ in plate noise by about 2–5× (plate-material ablation). Two independent reconstructions of the same eight scenes differed by up to 2.7 percentage points (e.g. `idli_pair__ceramic` -8.4 vs $-5.7\,\%$), which bounds run-to-run repeatability before the formal repeat experiment.

E. E5: Partly eaten plates, turntable path

Each of the six plates from E3 was paired with a leftover version on ceramic and steel: one or more items removed, the rest reduced in height and radius (e.g. the rice eaten and 60 % of the dal height left), rendered with the same seed, camera path and plate. Eaten volume is the difference of two

TABLE V
E5: EATEN VOLUME FROM BEFORE/AFTER 24-VIEW DENSE MVS ON PARTLY EATEN SYNTHETIC PLATES (EXACT MESH VOLUMES). SHARE: ERROR IN THE EATEN FRACTION OF THE SERVING, PERCENTAGE POINTS.

Plate	n	Median |err| % [95 % CI]	$\leq$10 %	Share (pp)
All	12	2.8 [2.0, 7.4]	11/12	2.8
Ceramic	6	7.4 [2.2, 9.8]	5/6	4.3
Steel	6	2.1 [1.5, 3.4]	6/6	2.1

TABLE VI
E6: EATEN SHARE OF THE SERVING FROM A 13-FRAME HEAD-MOUNTED SWEEP (SERVED AND CLEARED), SYNTHETIC PLATES. ERROR IN PERCENTAGE POINTS (PP).

Capture, footprint rule	n	Median |err| pp [95 % CI]	$\leq$10 pp	Max
Glasses, fixed rule	12	10.7 [5.2, 20.9]	5/12	106.1
Glasses, region growing	12	3.4 [0.7, 17.3]	8/12	37.5
Turntable (E5), fixed rule	12	2.8 [1.5, 5.4]	11/12	15.7

independent reconstructions, so the errors of both scans add. Across 12 pairs the median absolute error of the eaten volume is 2.8 % [2.0, 7.4], 11/12 within 10 %, and the eaten share of the serving is off by a median of 2.8 percentage points (Table V). The largest error is curry heap on ceramic (-11.1 %). The leftover scans alone are much worse in relative terms (median 9 % of the leftover volume), because thin remains on a plate are hard to reconstruct; in absolute terms that is a median 7.5 ml, small next to the served portion. These scans follow the 24-view turntable path of our first design; E6 repeats the same pairs from the glasses' viewpoint. Both are synthetic, not physical, measurements.

F. E6: Partly eaten plates seen from the glasses

Across 12 served/cleared pairs, the eaten share from the glasses sweep is off by a median of 10.7 percentage points with the fixed footprint rule used as the main rule in E5, and 5/12 pairs fall within 10 points (Table VI). The same plates on the turntable path give 2.8 points. The plate matters: on steel the median miss is 8.0 points (6 pairs), on white ceramic 16.5 points (6 pairs). The served scan alone is off by a median of +12.1 % in volume on steel and -25.2 % on ceramic, because a 60-degree sweep from one seat never sees the far side of the food, and, as in E3, white ceramic gives fewer features to match. Where the served and cleared scans are biased by different amounts, the bias does not cancel in the ratio. The largest miss is idli pair on ceramic (-106.1 points). The region-growing footprint, the second rule from E5, gives 3.4 points (8/12 within 10); we report it but did not choose it in advance. These are rendered scenes with exact volumes and a simulated head path, not measurements from the printed glasses.

G. E7: Which dish was left

E7 reuses the 12 E6 pairs. Three of the six plates hold more than one dish (a thali of rice, dal and curry; rice with dal; two idli), each on steel and on white ceramic, so there are 20 dishes, 14 of them sharing a plate. The truth for each

TABLE VII
E7: PER-DISH EATEN SHARE FROM THE CENTRE FRAME OF THE SERVED AND CLEARED BURSTS, ABSOLUTE ERROR IN PERCENTAGE POINTS (20 DISHES, 12 RENDERS).

	Median	95 % CI	$\leq$10	Max
All dishes	10.8	0.2–15.6	9/20	51.8
Steel	13.9	6.7–33.2	2/10	51.8
White ceramic	1.4	0.0–11.0	7/10	18.5
Area ratio only, no 1.5	24.3	0.8–28.0	7/20	54.9

dish is $1 - V_{\text{left}}/V_{\text{served}}$ from the scene meshes. Only the centre frame of each burst is used; nothing is reconstructed. Estimated items are paired with true dishes by position (Hungarian assignment). The colour threshold (12 Lab units), the texture threshold (6 grey levels above the rim), the 9-pixel vote window and the exponent 1.5 were fixed before the run. One change came after a first run: a unit test on drawn plates showed that the 7×7 texture window added a 3-pixel ring of bare plate around every dish and that this ring was handed to the wrong dish. The mask is now eroded by 3 pixels, and Table VII reports the run after that fix (script `analyze_items.py` in `experiments/synth`).

Without the exponent the median error is 24.3 points: the area a dish covers shrinks more slowly than its volume, so area alone says too little was eaten. The plate materials swap places compared with E6. White ceramic does well here and steel does badly, because the weak step is now the food mask rather than the 3D match. On steel some reflections on the bare plate pass as food. On ceramic, white rice is partly lost against the plate: the rice on the rice-and-dal plate covers 165 cm^2 in the steel mask and 19 cm^2 in the ceramic one. The worst dish is the dal next to that rice on steel, read as 10 % eaten when 62 % had gone.

The alert needs something coarser than the share: which dish was left. On all 6 plates with more than one dish, the dish with the lowest estimated share was also the one with the lowest true share. Four of the 6 dishes that were finished were read as at least 90 % eaten, and 3 of the 4 untouched dishes as at most 10 %. Six plates is a small sample and they are renders, so this shows that the split works on clean images, not that it is reliable on a real thali.

VIII. LIMITATIONS

Nutrition5k has no volume ground truth. The dataset provides overhead RealSense depth and weighed mass, not food volume. We therefore score monocular models against the volume integrated from sensor depth over the same dish region (food plus plate rim above the support plane). This measures how well a model reproduces a depth sensor, not how well it measures food. Our attempt to separate food from plate with a second plate-floor plane fit gave a food volume that barely tracks weighed mass (Spearman $\rho = 0.25$, Pearson $r = -0.02$, $n = 503$), so we do not use it as a target. Absolute food-volume accuracy is reported only on synthetic scenes with exact mesh volume.

Synthetic-to-real gap. Blender scenes have perfect camera intrinsics, controlled lighting and procedural textures. Real OV2640 frames add rolling shutter, auto-exposure drift, lens distortion and JPEG blocking on low-texture food. The resolution and JPEG ablations bound part of this gap but not all of it. Textureless food already fails in simulation (Sec. VII-C); an active texture projector or image-space segmentation would be needed for it.

No physical measurements yet. The glasses have not been printed, flashed or worn. Fit on a real temple arm, image sharpness during the head sweep, battery life, heat and Wi-Fi transfer time are untested, and the first turntable rig's bench tests have not been run either. None of the numbers in this paper come from hardware. Timings are CPU timings on 8-vCPU cloud VMs.

One seat, one side. A sweep from the carer's position sees the near side and top of the food, never the far side, so each volume is biased; the share is reliable only to the extent that the bias is the same in the served and the cleared scan (E6). A hand, cup or spoon in front of the plate, a missed button press or a plate moved between the two scans breaks the pair. The ArUco squares must lie beside the plate for metric scale; the share itself needs only a consistent scale between the two scans.

A share, not grams. In the earlier turntable design a load cell would have weighed the plate; the glasses cannot. They tell the carer that, say, 70 % of the serving went. To get grams, the kitchen would weigh each portion once when it is plated, and we would multiply. That only holds if what is left is as dense as what was served, which fails for a thali where the rice has gone and the dal is left.

No clinical or user study. The app, the alert rule and the weekly report have not been used by older adults, families or clinicians. Usability for people with poor eyesight or tremor, acceptance, alert fatigue and any effect on intake or outcomes are untested, and the residents in the app's demo data are fictional. A pilot in a ward or care home, with weighed plate waste as reference and staff and family feedback, is the next step.

The protein number rests on two guesses. First, the dish split. It sees one top view, so curd rice with sambar poured over it counts as one dish, an idli under a ladle of chutney is partly hidden, and turning the plate between the two presses breaks the pairing. The cluster names are not automatic yet either; E7 pairs clusters with dishes by where they sit. Second, the kitchen's protein figure, which is right only for a standard ladle. E7 has six plates with more than one dish, all rendered, and no dietitian or weighed plate waste has checked the alert.

IX. Conclusion

AssistVision turns a pair of ordinary spectacles into a meal-intake monitor for older adults: a clip-on camera records the plate when the meal is served and again when it is cleared, the host works out the share eaten, family members are alerted after a run of low-intake meals, and the doctor gets a weekly report. The parts cost about INR 2,000–2,900 and two of them are printed. The vision experiments explain the design. A single image anchored only on the table does not recover food volume: the best deployable models (ZoeDepth-NYU, and Depth Pro at 15× the latency) still have median errors of 44–48 %, and low depth error (AbsRel) does not imply low volume error. Several views with marker scale work on textured synthetic food on a turntable path (E5). From the glasses' short head sweep (E6) the share eaten is off by a median of 8 points on steel and 16.5 on white ceramic, against 2.8 on the turntable path, and only 5 of 12 pairs come within 10 points. Textureless food fails in both cases, and volume does not predict grams on real mixed dishes, which is why the glasses report a share rather than grams. The same two photos also show which dish was left: on 6 simulated multi-dish plates the dish left most was named correctly each time, and the app turns that into a protein-gap alert that says whether the dal is being left or the menu is short. Next we will print and wear the glasses, capture real thalis against weighed plate waste, name the dishes automatically, and pilot the app with residents, families and staff in a care home.

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Parts of the software and a first draft of this text were prepared with an AI coding assistant (Devin, Cognition AI). The authors checked, edited and take responsibility for all content, code and results.

Reproducibility

All code, scene specifications, per-dish result files and the scripts that generate every table and figure are in the repository. Evidence categories are labelled throughout: *Nutrition5k* (public real RGB-D data), *synthetic* (Blender/Cycles renders with exact mesh volume), *VM* (timings on an 8-core CPU-only cloud VM, not on the ESP32), and *physical* (pending hardware measurements).

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